

Unit 12, Lesson 1: Finding Surface Area of Right Circular Cylinders Using Nets

Benchmark

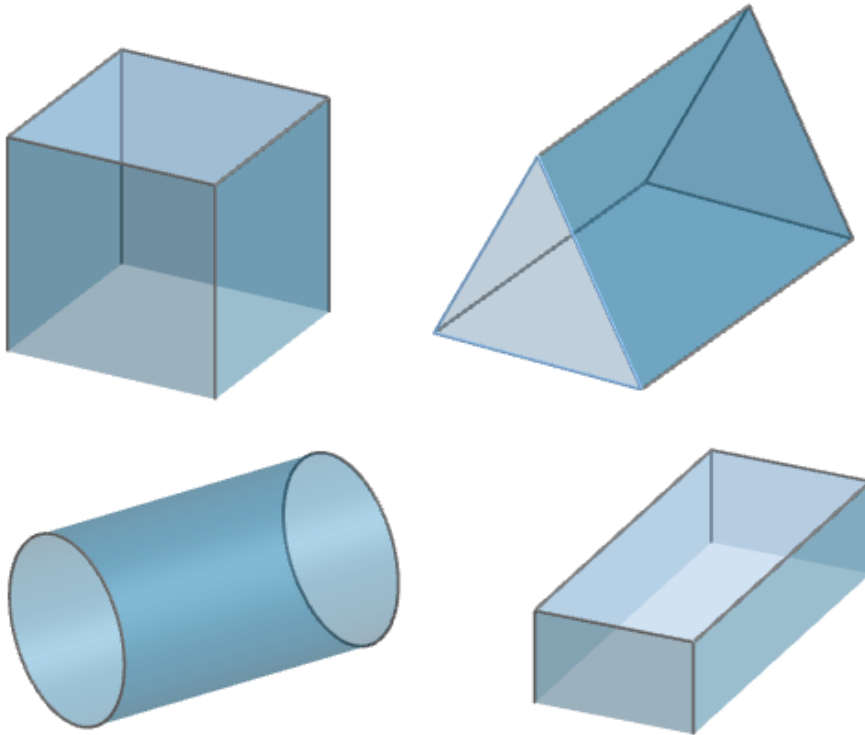
MA.7.GR.2.1 Given a mathematical or real-world context, find the surface area of a right circular cylinder using the figure's net.

Learning Targets

- ☐ I can draw the net of a right circular cylinder.
- ☐ I can find the surface area of a right circular cylinder using the figure's net.

12.1.1 Warm-Up: Which One Doesn't Belong?

1. Circle the figure that does not belong.

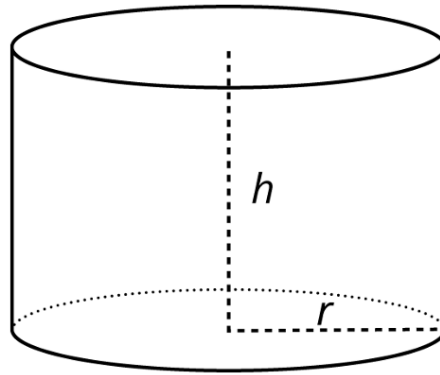


2. Explain why that figure does not belong.

12.1.2 Guided Instruction: Drawing the Net of a Right Circular Cylinder

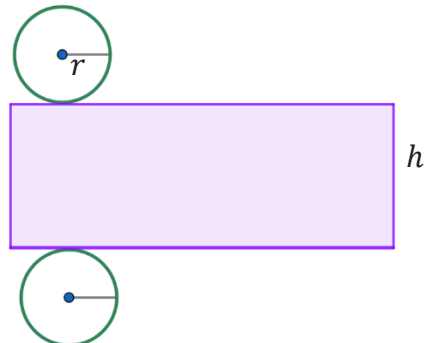
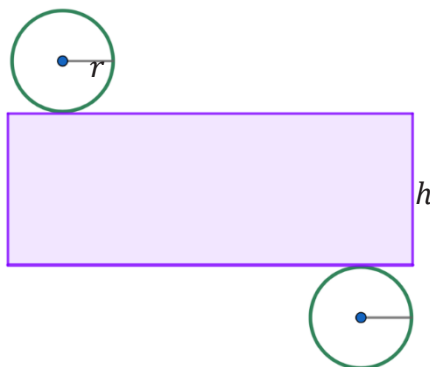
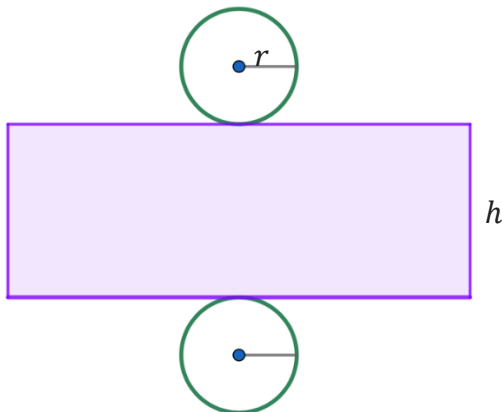


A **cylinder** is a figure containing two congruent, parallel, circular bases whose edges are connected by a perpendicular curved surface. The net of a cylinder consists of a parallelogram and two circles.



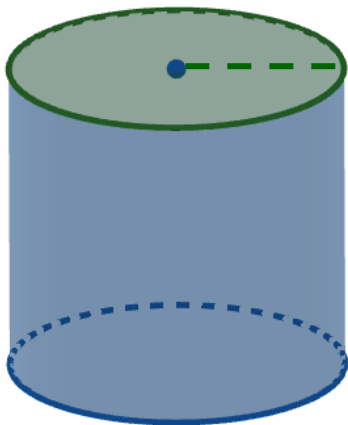
A net is a two-dimensional representation of a three-dimensional figure that can be folded to form the three-dimensional shape.

- Discuss with a partner if these nets can be folded into a right circular cylinder. Circle the nets that represent a net of a right circular cylinder.



2. Draw the net of the right circular cylinder shown. Label the net with its dimensions. Then, using the terms in the word bank, label its parts.

$$r = 4 \text{ centimeters (cm.)}$$



$$h = 9 \text{ cm.}$$

Parts of a Right Cylinder
Circular base
Curved surface

3. Tin cans are used for packaging many things, including various types of foods. The tin can shown has a diameter of 4 inches (in.) and a height of 6 in. Draw the net of the tin can. Label the net with its dimensions.

$$d = 4 \text{ in.}$$



$$h = 6 \text{ in.}$$

12.1.3 Exploration: Relationship Between Circumference and Lateral Area of a Right Circular Cylinder



The perpendicular curved surface that connects the circular bases of the right circular cylinder is also called the ***lateral area*** of the cylinder.

1. The diagrams show a circle with a diameter of 4.9 units. The circle is then rolled along the ruler so that the circumference (represented in green) is “peeled” off.

Diagram 1

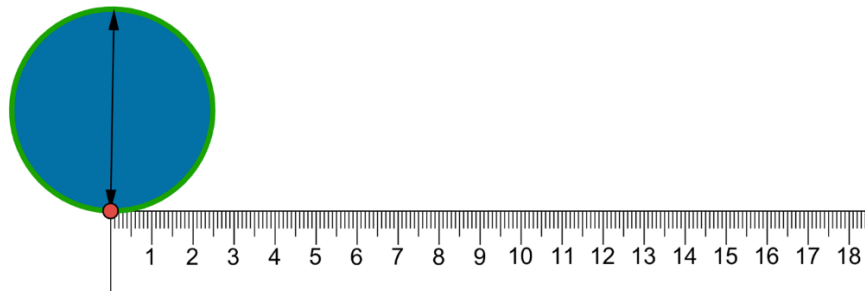


Diagram 2

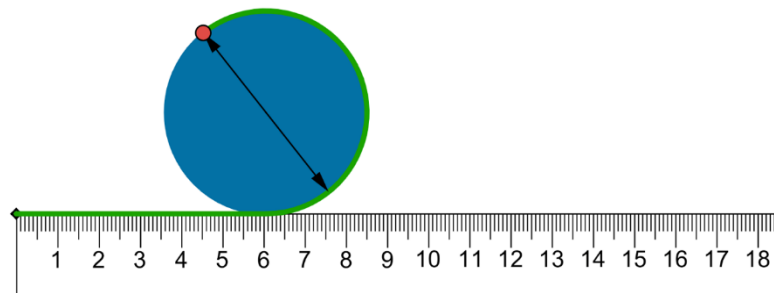
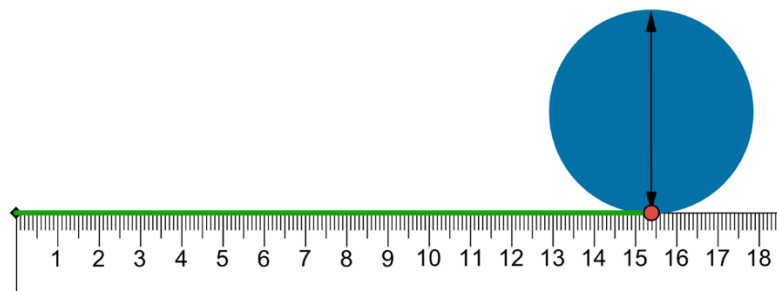


Diagram 3



If the circle represents a circular base of a right circular cylinder, discuss with your partner how the diagrams show the relationship between the circumference of the circle and the length of the lateral area of the cylinder. Write a summary of the discussion.

2. Willard's grandmother collects vintage labels from canned goods. The label shown is from a can of squash.



- a. Discuss with your partner how the label relates to the net of a cylinder. Draw a net of the can, labeling the part of the net that represents the vintage label.
- b. Complete the statement to describe how the length of the label is related to the circular base when the label is wrapped around the can.

The length of the label is equal to the _____
of the circular base.

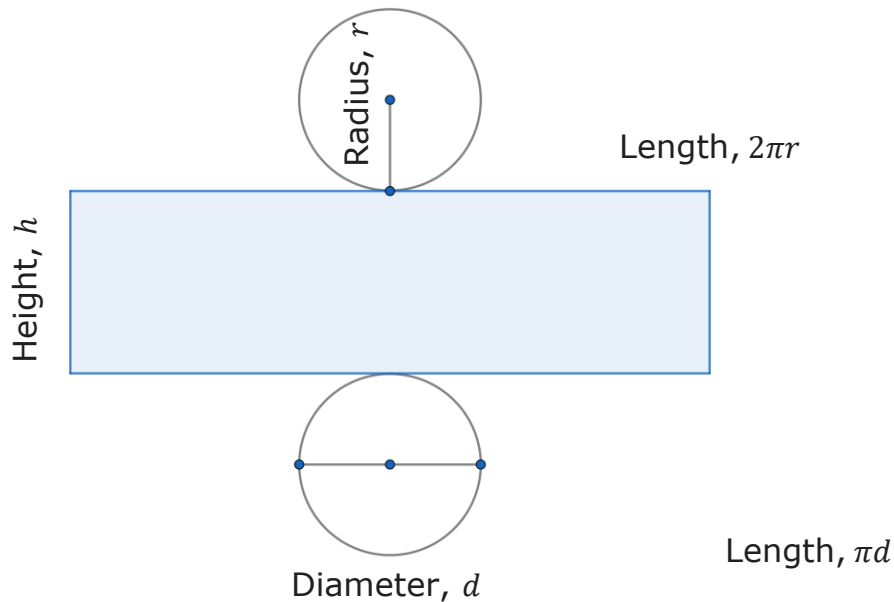
- ☐ area
- ☐ circumference
- ☐ diameter
- ☐ radius

12.1.4 Guided Instruction: Surface Area of a Right Circular Cylinder



The **surface area of a right cylinder** is the total area of the two congruent, parallel circular bases and the rectangular curved surface. Surface area is measured in square units.

1. A net of a right circular cylinder is shown with its labeled dimensions.



- a. Discuss with your partner why there are two labels for the length of the rectangle.
- b. Select all of the formulas that can be used to find the area of 1 of the circular bases and the area of the rectangle.

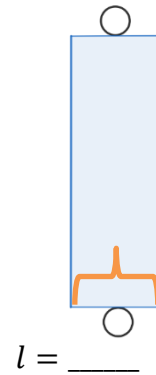
Area of 1 of the Circular Bases		Area of the Rectangle	
☐ πd^2	☐ $2\pi r \cdot h$	☐ $2 \cdot \pi \cdot d$	☐ πr^2
☐ πr^2	☐ $\pi \left(\frac{1}{2}d\right)^2$	☐ $2\pi r \cdot h$	☐ $2\pi r$
☐ $\pi r \cdot h$	☐ $2\pi r$	☐ πd	☐ $h \cdot \pi \cdot d$

2. For an art project, Caleb will use a new dowel that measures 2 inches (in.) in diameter and 19 in. in height. To find the surface area of the dowel, Caleb drew the net below. Use 3.14 as π .

3.



$$h = 19 \text{ in.}$$

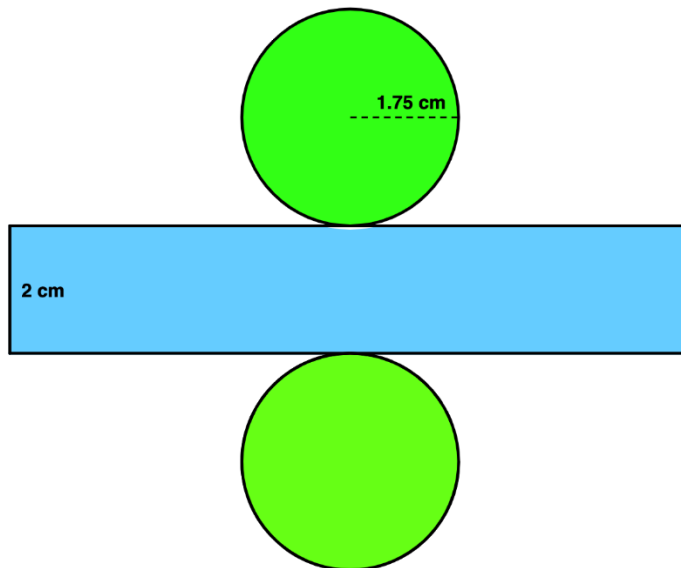


Complete the following for the given context. Represent your answers as an approximation by using 3.14 for π .

- Find the missing value in the net Caleb drew. Include units.
- What is the area of one of the circular bases? Include units.
- What is the circumference of one of the circular bases? Include units.
- Before Caleb can use the new dowel, he must first remove the tight plastic casing the entire dowel is enclosed in. How much plastic is the dowel enclosed in? Include units.

12.1.5 Wrap-Up: Ticket Out the Door

1. The net of a right circular cylinder is shown.



Use the net to find the surface area of the right circular cylinder. Use $\frac{22}{7}$ for π and include units.

